MIL-M-17321F
30 September 1985
SUPERSEDING
MIL-M-17321E
19 June 1973

MILITARY SPECIFICATION

MARKERS, TRAFFIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and two sizes of traffic markers used in the maintenance of traffic control systems.

1.2 Classification. Markers shall be in the following sizes, as specified (see 6.2):

Size 1 - 18 inches
Size 2 - 28 inches

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9905

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Manual on Uniform Traffic Control Devices for Streets and Highways

(Application for copies should be addressed to the Superintendant of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Highway Orange Color Tolerance Chart

(Application for copies should be addressed to the U.S. Department of Transportation, Federal Highway Administration, Washington, DC 20591.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Tension Testing of Vulcanized Rubber

D 573 - Accelerated Aging of Vulcanized Rubber by the Oven Method

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D 2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 Material. Traffic markers shall be made of either plastic or rubber. Recycled material may be used (see 6.3).

* 3.1.1 Plastic. Plastic material used for the manufacture of traffic markers shall be virgin polyvinyl chloride resin and so formulated as to meet the physical properties in table I when tested as specified in 4.2.3.

TABLE I. Physical properties of plastic

Property	Requirement
Minimum tensile strength, in psi	1200
Minimum ultimate elongation, in percent	200
Hardness of Shore A durometer, in points:	
Conical section	70 ± 10
Base	80 ± 10

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* 3.1.1.1 Color. The color for the outer layer of the conical upper portion shall be a fluorescent red-orange. The inner layer may be translucent or white pigmented. The base shall be either black pigmented or white pigmented, or fluorescent red-orange. The color for the plastic cones shall permeate throughout the outer layer rather than appear only on its exterior surface.

3.1.1.2 Heat resistance. The marker shall support itself and show no evidence of sagging when tested as specified in 4.2.3.

3.1.1.3 Low temperature resistance. The marker shall show no evidence of cracking when tested as specified in 4.2.3.

3.1.1.4 Design and construction. The plastic marker shall conform to the design, shape, and dimensions as shown in figure 1. The marker shall be of single piece construction, having a base whose color is specified in 3.1.1.1 fused to a fluorescent conical section. The fluorescent conical section shall consist of a white underlayer fused to a fluorescent outerlayer.

3.1.2 Rubber. Rubber material used for the manufacture of markers shall be natural rubber, reclaimed rubber, synthetic rubber (styrene-butadiene), or blends thereof, plus compounding ingredients of such nature and quality as to produce material meeting the physical requirements of table II when tested as specified in 4.2.3.

TABLE II. Physical properties of rubber

Property	Requirement
Minimum tensile strength, in psi	1000
Minimum ultimate elongation, in percent	200
Hardness of Shore A durometer, in points	70 ± 5

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3.1.2.1 Color. The color of the rubber marker shall be as shown on figure 2. The orange color shall match the standards for Highway Orange, PR Color No. 6, as shown on Highway Orange Color Tolerance Chart.

3.1.2.2 Accelerated aging. When tested as specified in 4.2.3, change in physical properties shall not exceed the following:

Tensile strength	25 Percent
Ultimate elongation	25 Percent
Hardness	10 Points

3.1.2.3 Preparation of surface prior to painting. The area to be painted, as specified on the figures, shall be cleaned by wire brushing prior to painting.

3.1.2.3.1 Paint adhesion. The paint shall show no cracking when tested as specified in 4.2.3.

3.1.2.4 Design and construction. The rubber markers shall conform to the design, shape, and dimensions as shown on figure 2 and as specified herein. The markers shall be of single piece construction.

3.2 Stacking. Traffic markers shall have a pattern of equally positioned spacers and feet as shown on the figures to prevent sticking while stored in a stacked position.

3.3 Identification marking. All traffic markers shall be marked with the year of manufacture and contractor's name or trademark so as to be readily identified with the contractor. The method of application shall ensure that the lettering cannot be removed without damaging the markers.

3.4 Workmanship. Traffic markers shall be free of burns and discoloration. Walls of cones shall be smooth and free of bumps, ridges, or voids. Markers shall be free from blistered, peeled, cracked, or chipped paint. The occurrence of defects shall not exceed the acceptable quality levels (AQLs) specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. The contractor shall furnish a certificate of compliance stating that he has complied with the requirement in 3.1.2.3. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.2.2 Inspection of the end item. The end item shall be examined in accordance with 4.2.2.1 and 4.2.2.2 and tested in accordance with 4.2.3. Random samples shall be drawn from each lot of the end items for examination of visual, dimensional, and packaging defects. The lot shall consist of all traffic markers submitted for inspection at one time. The lot, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of markers for examination.

4.2.2.1 Examination for visual defects. Examination shall be made for the defects listed in table III. The sample unit for this examination shall be one marker. The inspection level shall be II and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE III. Visual examination

Examine	Defect
Workmanship	Contains burns or discoloration Walls are not smooth; contain bumps, ridges or voids Blistered, peeled, chipped
Color	Not as specified Orange peel, smudge, sag Foreign matter imbedded
Design and construction	Not as specified
Marking	Not as specified

4.2.2.2 Examination for defects in dimensions. The sample unit for this examination shall be one marker. Wall thickness shall be measured at 5 locations, 1 inch up from the base and 1 inch down from the top. The average of the total five shall not be less than the required minimum. Any deviation from the specified dimensions shall be classified as a defect. The inspection level shall be S-3 and the AQL shall be 2.5, expressed in terms of percent defective.

4.2.3 Testing of the end item. Wherever applicable, testing shall be done in accordance with ASTM Methods. The item shall be tested for applicable characterstics listed in table IV. The lot size shall be expressed in units of markers. The sample unit shall be one marker. Failure of one or more sample units to meet individual unit requirements shall be cause for rejection of the lot.

Lot size (markers)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE IV. Instructions for testing

Characteristic	Document Reference		Requirement Applicable to Individual Unit	Number Determinations per Sample Unit	Reported Results	
	Requirement	Test Method			Pass or Fail	Numerically to Nearest
Plastic:						
Material	1/					
Tensile strength	Table 1	D 412	X	2		1 psi
Ultimate elongation	Table 1	D 412	X	2		1%
Hardness of						
Conical section	Table I	D 2240	X	2		1 point
Base	Table I	D 2240	X	2		1 point
Weight						
Total	Figure 1	4.3.1	X	2		1 oz
Base	Figure 1	4.3.1	X	2		1 oz
Heat Resistance	3.1.1.2	4.3.3	X	2	X	
Low temperature resistance	3.1.1.3	4.3.4	X	2	X	
Rubber:						
Material	1/					
Tensile strength	Table II	D 412	X	2		1 psi
Ultimate elongation	Table II	D 412	X	2		1%
Hardness	Table II	D 2240	X	2		1 point
Weight	Figure 2	4.3.1	X	2		1 oz
Accelerated aging	3.1.2.2	4.3.5	X	2		as applicable
Paint adhesion	3.1.2.3.1	4.3.2	X	1	X	

1/ A certificate of compliance shall be furnished to the contracting officer by the contractor.

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4.2.4 Packaging inspection. An examination shall be made to determine that the packaging, preservation, packing, and marking complies with the requirements in section 5. Defects shall be scored in accordance with table V. The lot size shall be the number of shipping containers in the end item inspection lot. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE V. Packaging inspection

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method application
Materials	Any component missing or not as specified Any component damaged affecting serviceability
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping Bulged or distorted container
Content	Markers per container exceed net weight limitation

* 4.2.5 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table VI. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE VI. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3 Test methods.

4.3.1 Weight. Any balance accurate to one ounce shall be satisfactory for this determination.

4.3.2 Paint adhesion. The paint shall be tested for adhesion to the rubber cone 48 hours after drying. The painted area shall be squeezed together at the top, middle, and bottom and held for 5 seconds and released. Paint shall be examined for cracking and lifting.

4.3.3 Heat resistance. The plastic traffic cones shall be placed in an air-circulating oven at a temperature of 150 $\pm 2^{\circ}$F for a period of two hours. Upon removal from the oven, the test cone shall show no evidence of sagging.

4.3.4 Low temperature resistance. The plastic traffic markers shall be placed in a cold-box at -10°F for a period of two hours. Upon removal of the marker from the cold-box, the conical portion shall be bent 180 degrees at a point near the middle of its vertical height. The marker shall show no evidence of cracking.

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4.3.5 Accelerated aging. Aging shall be determined in accordance with ASTM D 573. The aging period shall be 70 hours at 158°F. The aging test of the rubber compound may be done on the molded cone or molded panels. Specimens for testing will be cut from either the marker or molded panel.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The markers shall be nested. Before nesting, the markers shall be wrapped with a suitable material to prevent the removal of paint and to prevent adhesion of the markers to each other.

* 5.1.2 Commercial. The markers shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Markers of one size only, preserved as specifed in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

* 5.2.2 Level B packing. Markers of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Markers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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* 5.3 Palletization. When specified (see 6.2), markers packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the patterns of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be first approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with the requirements of MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The traffic markers are intended for use in traffic channeling utility repairs, line painting, and emergency blockades. The 18-inch markers are used primarily when painting traffic lines in order to keep cars off the wet paint. The 28-inch markers are used primarily for blocking off any type of area where construction work is being done.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. Size required (see 1.2).

c. Selection of applicable level of packaging and packing (see 5.1 and 5.2).

d. When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.1).

e. When palletization is required (see 5.3).

* 6.3 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.1).

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6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only and the Government assumes no liability whatsoever for inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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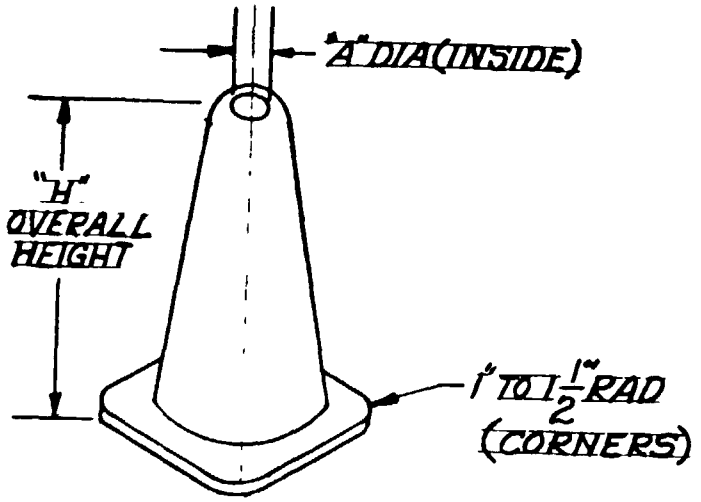
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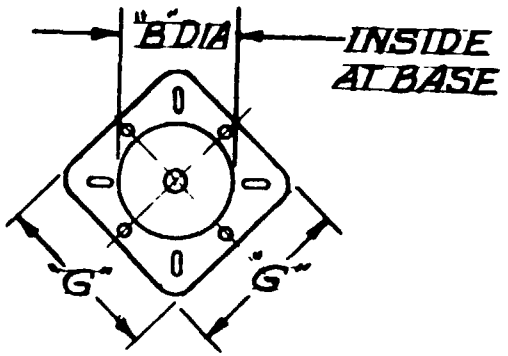
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BASE
BOTTOM VIEW SHOWING FEET AND SPACERS

NOM SIZE	SQUARE BASE "G" (MIN)	BASE WT (MIN)	OVERALL HGT "H"	TOP WALL THICK	BOTTOM WALL THICK	TOP I.D. "A" (MIN)	BOT I.D. "B" (MIN)	TOTAL CONE WT (MIN)
18	11 1/4"	2LB-2 OZ	18" MIN	.080" MIN	.090" MIN	1 7/32"	7 7/16"	3 LB-8 OZ
28	14"	4 LB-4 OZ	28" +1/2 -1/2	.090" MIN	.100" MIN	1 1/4"	10 5/8"	7 LB-4 OZ

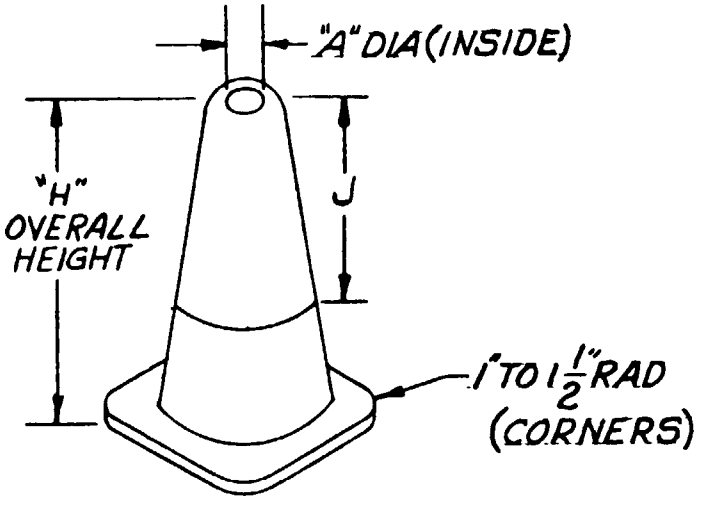
FIGURE 1: PLASTICS (POLYVINYL CHLORIDE) TRAFFIC MARKERS

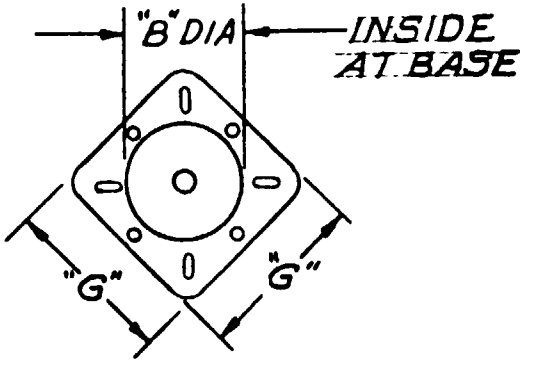
NOTES:

1. FEET AND/OR SPACERS AS REQUIRED TO PREVENT STICKING IN STACKED POSITION (HEIGHT TO BE A MINIMUM OF 1/4 INCH).
2. INSIDE DIAMETER AT TOP OF CONE TO BE OF SUCH DESIGN SO AS TO ACCEPT SIGN OR FLAG ADAPTER.

Figure 1. Plastic (Polyvinyl Chloride) Traffic Markers

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BASE
BOTTOM VIEW SHOWING FEET AND SPACERS

NOM SIZE	SQUARE BASE "G" (MIN)	OVERALL HGT "H"	TOP WALL THK (MIN)	BOTTOM WALL THICK (MIN)	TOP I.D. "A" (MIN)	BOT I.D. "B" (MIN)	TOTAL CONE WT (MIN)	ORANGE PAINT "J" HGT
18	11 1/4"	18" MIN	1/8"	3/16"	1 23/32"	7 3/4"	3 LB	13" + 1/2 - 1/2
28	14"	28" + 1/2 - 1/2	5/32"	3/16"	1 23/32"	10 3/4"	7 LB	22 3/4" + 1/2 - 1/2

FIGURE 2: RUBBER TRAFFIC MARKERS

NOTES:

1. FEET AND/OR SPACERS AS REQUIRED TO PREVENT STICKING IN STACKED POSITION (HEIGHT TO BE A MINIMUM OF 1/4 INCH).
2. INSIDE DIAMETER AT TOP OF CONE TO BE OF SUCH DESIGN SO AS TO ACCEPT SIGN OR FLAG ADAPTER.

Figure 2. Rubber Traffic Markers

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-17321F	Markers, Traffic

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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